Course 2131 | Semester II | Theory | 4 Credits

Fundamentals of Electrical and Electronics Engineering

A full bilingual student handbook for circuit laws, domestic electrical installations, passive components, semiconductors and digital electronics.

Programmes	**FT&S · Mechatronics · T&D · W&P**
Contact hours	**60**
Teaching scheme	**4:0:0:0**
CIE / ESE	**40 / 60**
Modules	**4 × 15 h**
Self-learning	**6 h/week; 90 h listed**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

This handbook is based only on the supplied official **Diploma Curriculum Revision 2026** syllabus PDF for Course **2131**, titled **Fundamentals of Electrical and Electronics Engineering**. No Revision 2021 course content has been merged.

Exact official identity: Semester II, Theory, 60 contact hours, 4 credits, teaching scheme 4:0:0:0, CIE 40 and ESE 60. The official programme row lists Fire Technology and Safety, Mechatronics, Tool & Die Engineering, and Wood and Paper Technology.

Absent official fields: A separate equipment list, laboratory specification and practical examination are not specified in the supplied PDF. This is a theory course.

Verified source issues retained transparently

PDF location / wording	Issue	Handbook treatment
Detailed syllabus, Module III — Capacitors: "Fixed resistors, Variable and Special purpose capacitors".	"Fixed resistors" is clearly inconsistent with the capacitor heading.	The exact wording is disclosed here; teaching uses the technically coherent categories fixed, variable and special-purpose capacitors.
Detailed syllabus, Module IV maps PN diode, rectifiers and BJT to CO4; CO4 wording mentions digital electronics, number systems and logic gates.	CO mapping and CO wording do not fully align.	Official mapping is preserved as CO4 in the coverage table. Explanations do not rewrite the official CO.
Module I detailed syllabus names Fleming's Right-Hand Rule for AC generation. Self-learning Weeks 4 and 5 both ask for Fleming's Left-Hand Rule.	Rule mismatch and duplicated activity across Modules I and II.	Both official activities are reproduced. The handbook clearly distinguishes generator Right-Hand Rule from motor Left-Hand Rule.
Teaching scheme shows self-learning 6 h/week; the activity table lists 15 activities × 6 h and explicitly totals 90 h.	Not a numerical conflict: 15 weeks × 6 h = 90 h, but it is separate from 60 contact hours.	Dashboard displays 60 contact hours and 90 listed self-learning hours separately.
Capacitor and inductor specification rows include voltage, current and power ratings.	These are official terms, though common datasheets emphasize different primary ratings depending on component.	Official terms are retained, with a note on practical datasheet interpretation.

COURSE DASHBOARD

What You Are Expected to Learn

60 h

Official contact hours

4
Credits

40 + 60
CIE + ESE marks

90 h
Listed self-learning

Course introduction

The course builds basic knowledge of electrical circuits, domestic supply, safety, energy use, electronic components, semiconductor devices, number systems and logic gates. Its purpose is foundational engineering practice across multiple programmes.

> **Malayalam support:** □ course-□ Electrical circuit □□□□ Domestic wiring, Electronic components, Diode, Transistor, Number system, Logic gate □□□ □□□□□□□□□□□□□ □□□□□□□□□□□□. □□□ topic-□□ practical engineering □□□□□□□□□□□□□□ □□□□□□□□□□□□□□□□□ □□□□□□□□□.

Prerequisites

- **1002 — Fundamentals of Engineering Mathematics I** (Semester I)
- **2002B — Engineering Physics for Applied Electrical Technology and Computing I** (Semester I)

The PDF lists prerequisite topics as “Fundamental of Mathematics” and “Basic knowledge in Physics”.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Total contact hours	Credits	Theory CIE	Theory ESE	Total
4	0	0	0	6 h	60	4	40	60	100

STUDY METHOD

How to Use This Handbook

1 · Understand

Read the concept and identify symbols, units and physical meaning.

2 · Visualise

Study the circuit, waveform, block diagram or component drawing.

3 · Apply

Follow each worked example, then solve the self-check without looking at the answer.

4 · Revise

Use formula banks, common mistakes and module questions before examinations.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. To provide fundamental knowledge of electrical engineering concepts and basic circuit laws for understanding simple electrical systems.
2. To familiarize students with domestic electrical supply systems, wiring practices, safety measures, and energy utilization in everyday applications.
3. To enable students to identify and understand passive and active electronic components and their role in basic electronic circuits.
4. To introduce the basics of digital electronics, including number systems and logic gates, for developing foundational analytical and problem-solving skills.

Student-friendly meaning

You should be able to calculate simple circuit values, recognise household supply and protection, identify major electronic components, explain diode/transistor basics, convert number systems and use truth tables.

> **Malayalam support:** □□□□□□□□□□□ □□□□□□□ simple circuit calculation □□□□□□□□□□, □□□□□□□□□ supply/protection arrangement □□□□□□□□□□□□□□□□, common electronic

components identify □□□□□□□□□, number conversion, logic gate truth table □□□□□□□□□□□□□□ □□□□□□□.

Official Course Outcomes

CO	Official wording	Bloom level
CO1	Apply fundamental electrical concepts and basic circuit laws to study simple DC and AC electrical circuits.	Apply
CO2	Use the knowledge of domestic electrical supply systems, wiring practices, safety devices, and energy utilization to study household electrical installations.	Apply
CO3	Illustrate the principles of passive and active electronic components to study basic electronic circuits and devices.	Apply
CO4	Understand the fundamentals of digital electronics, including number systems, number conversions, and basic logic gates.	Understand

Official CO–PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	3	–	–
CO2	3	2	2	2	3	2	–
CO3	3	2	2	2	3	–	–
CO4	2	2	–	–	2	–	–
Total	**11**	**8**	**6**	**6**	**11**	**2**	**–**
Correlation	**2.67**	**2**	**2**	**2**	**2.67**	**2**	**–**

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Official topics	Hours	CO	Cognitive levels in detailed syllabus	Handbook section
I	Electricity history; current, voltage, resistance; AC/DC; Ohm's law; law and combinations of resistance; voltage/current division; Faraday laws; AC generation; sinusoidal parameters.	15	CO1	Remember, Understand, Apply	Fundamental Laws
II	Circuit terms; domestic supply structure; protection; earthing; safety and shock; residential wiring; energy consumption.	15	CO2	Remember, Understand, Apply	Domestic Supply & Energy
III	Evolution; resistor and colour code; capacitor and number code; inductor; transformer.	15	CO3	Understand, Apply	Passive Components
IV	PN diode; rectifiers; BJT; digital electronics; number systems; logic gates; IoT and AI.	15	CO4 (official mapping)	Remember, Understand, Apply	Semiconductor & Digital

LEARNING ROADMAP

How the Four Modules Connect

Electrical quantities

Voltage drives current through resistance.

Safe utilization

Supply, protection, wiring and energy measurement.

Building blocks

R, C, L and transformer shape electrical behaviour.

Electronic decisions

Diodes, transistors and logic gates process power and information.

MODULE I

Fundamental Laws in Electrical Engineering

From basic electrical quantities to resistance networks, electromagnetic induction and sinusoidal AC.

15 hours **CO1** **Remember · Understand · Apply**

Official Module I subtopic allocation

Subtopic	Hours	Taxonomy
Introduction to electricity	1	Remember
Current, voltage and resistance	1	Remember
Concept of AC and DC	1	Understand
Ohm’s Law	1.5	Apply
Law of resistance	0.5	Apply
Combination of resistance	3	Apply
Voltage and current division rules	3	Apply
Faraday’s laws	1	Understand
Generation of AC voltage	2	Understand
Sinusoidal representation and parameters	1	Understand
Total	**15**	**CO1**

Learning goals

- Define current, voltage and resistance with SI units.
- Solve series/parallel networks and division rules.
- Explain induction and AC generation.

Key instruments

Ammeter is connected in series; voltmeter in parallel; ohmmeter only on a de-energized circuit.

Engineering habit

Always draw polarity, current direction and units before calculation.

1.1 Introduction to electricity — milestones and physical idea−

Electricity is the set of phenomena associated with electric charge. Important milestones include observations of static charge, Volta's chemical cell, Ørsted's link between current and magnetism, Faraday's induction, and the later development of generators, transformers, semiconductor devices and integrated electronics.

In metallic conductors, free electrons drift when an electric field is applied. Conventional current direction is defined from positive potential to negative potential, opposite to electron drift.

> **Malayalam support:** Metal conductor-□ electrons move □□□□□□□□□□□□□□□ circuit diagram-□ **conventional current** positive terminal □□□□ negative terminal □□□ □□□□□ □□□□□□□□□□□□. Electron flow □□□□□□□ □□□□□□□ □□□□□□□□□.

1.2 Current, voltage and resistance−

Current

Rate of flow of charge.

$$I = Q / t$$

I in ampere (A), **Q** in coulomb (C), **t** in second (s).

Voltage

Work done or energy transferred per unit charge.

$$V = W / Q$$

V in volt (V), **W** in joule (J).

Resistance

Opposition offered to current.

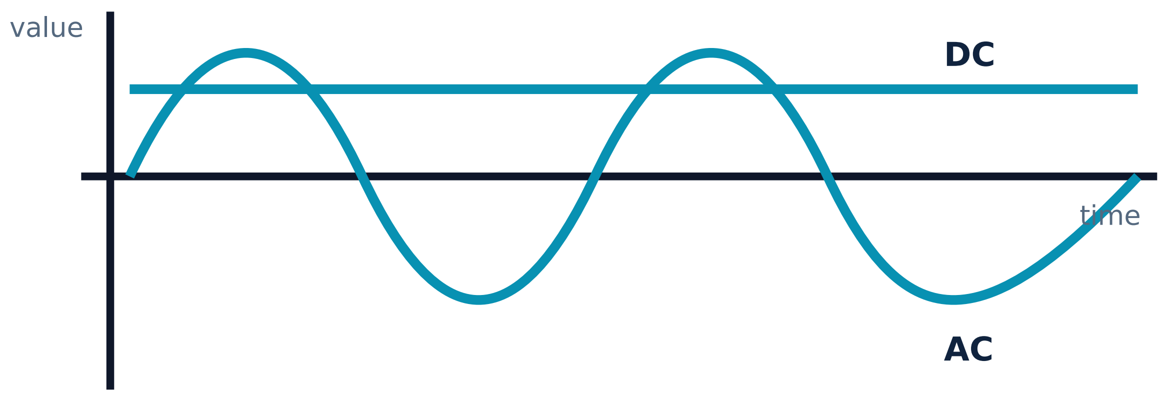
$$R = V / I$$

R in ohm (Ω).

Malayalam support: Voltage □□□ “electrical pressure” □□□□□□□, Current charge flow rate □□□□, Resistance □ flow-□□ □□□□□□□ □□□□□□□□ □□□□□□□□□□□□□□□□. □□□ analogy □□□□□□; exact physical definition formula-□□□□□□□.

1.3 AC and DC — waveforms, sources and applications−

DC maintains one polarity; its magnitude may be steady or pulsating. Batteries, DC supplies and solar cells are common sources. **AC** reverses polarity periodically; utility supply and alternators are common sources.



DC keeps one polarity; sinusoidal AC alternates positive and negative.

Typical uses

- AC: grid supply, induction motors, transformers.
- DC: electronic circuits, battery systems, control circuits.

Common mistake: “DC is always constant.” Pulsating DC keeps polarity but can vary in magnitude.

1.4 Ohm’s Law and electrical power−

For a conductor whose physical conditions remain constant, current is directly proportional to applied voltage.

$$V = IR \quad | \quad I = V/R \quad | \quad R = V/I$$

Electrical power is the rate of energy conversion:

$$P = VI = I^2R = V^2/R$$

Worked example — lamp current

Problem: A 24 V lamp has resistance 12 Ω. Find current and power.

Given: V = 24 V, R = 12 Ω. **Required:** I and P.

Calculation: I = V/R = 24/12 = 2 A. P = VI = 24 × 2 = 48 W.

Answer: Current = 2 A; power = 48 W.

Malayalam support: Ohm’s Law □□□□□□□□□□□□□□□□ V, I, R □□□□□ given, □□□□□ required □□□□□ □□□□□ □□□□□□. Unit □□□□□□□□ final answer exam-□ □□□□□□□□□□□.

1.5 Law of resistance and resistivity−

$$R = \rho L/A$$

ρ is resistivity (Ω·m), **L** length (m), and **A** cross-sectional area (m^2). Resistance increases with length, decreases with area and depends on material and temperature.

Worked example — wire length

A wire has $\rho = 1.7 \times 10^{-8}$ Ω·m, L = 10 m and A = 1 mm^2 = 1×10^{-6} m^2.

$R = (1.7 \times 10^{-8} \times 10)/(1 \times 10^{-6}) = 0.17$ Ω.

Resistance = 0.17 Ω.

1.6 Series, parallel and series-parallel resistance–

Series

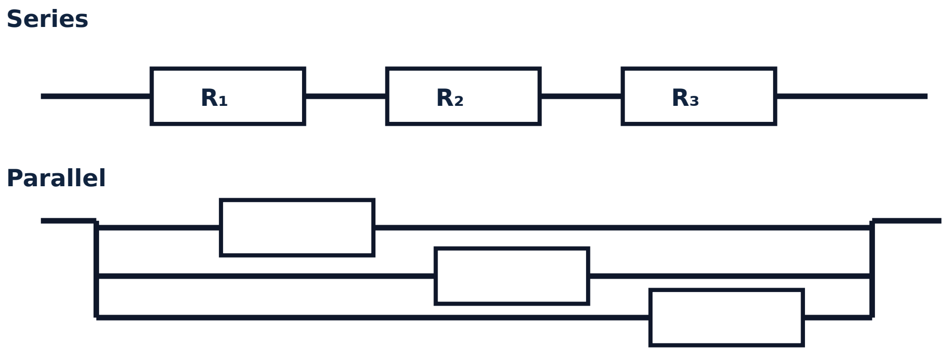
$$R_T = R_1 + R_2 + \ldots$$

Same current flows through every resistor. Voltages add.

Parallel

$$1/R_T = 1/R_1 + 1/R_2 + \ldots$$

Same voltage appears across every branch. Branch currents add.



Series: one current path. Parallel: multiple branches between the same two nodes.

Worked example — series-parallel

$R_1 = 4\ \Omega$ is in series with $R_2 = 6\ \Omega$ and $R_3 = 3\ \Omega$ in parallel.

$R_2 \| R_3 = (6\times3)/(6+3) = 2\ \Omega$. Therefore $R_T = 4 + 2 = 6\ \Omega$.

At 12 V, total current $I = 12/6 = 2$ A.

Equivalent resistance = 6 Ω; source current = 2 A.

1.7 Voltage Division and Current Division Rules–

Voltage division

$$V_k = V_T \times R_k/R_T$$

Applicable to series resistors.

Current division — two branches

$$I_1 = I_T \times R_2/(R_1+R_2)$$

Current divides inversely with branch resistance.

Worked example — divider

Two series resistors 2 kΩ and 3 kΩ are connected to 10 V.

Voltage across 3 kΩ = 10 × 3/(2+3) = 6 V. Voltage across 2 kΩ = 4 V. Check: 4 + 6 = 10 V.

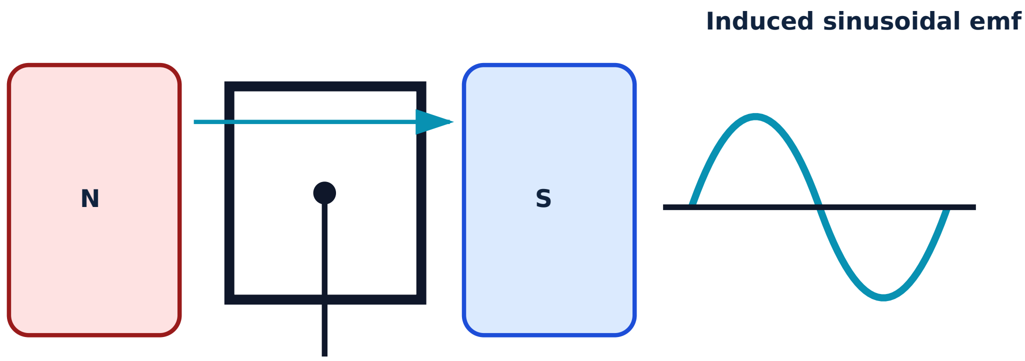
Common mistake: Current division uses the *other* branch resistance in the numerator for a two-branch network.

1.8 Faraday’s laws, Lenz’s law and AC generation−

Faraday’s First Law: An emf is induced whenever magnetic flux linking a circuit changes. **Second Law:** Magnitude of induced emf is proportional to rate of change of flux linkage.

$$e = -N\, d\Phi/dt$$

The negative sign represents **Lenz’s Law**: induced action opposes the change causing it. Fleming’s **Right-Hand Rule** gives induced current direction in a generator. Fleming’s **Left-Hand Rule** gives force direction in a motor.



Rotating a coil changes flux linkage and produces alternating emf. The animation is conceptual, not a construction drawing.

Malayalam support: Generator-□ conductor magnetic field cut □□□□□□□□□□□ emf induce □□□□□□□. Direction □□□□□□□□□□ Right-Hand Rule; motor force direction □□□□□□□□□□ Left-Hand Rule. □ □□□□□□ □□□□□□□□□.

1.9 Sinusoidal waveform and parameters−

$v(t) = V_m \sin(\omega t + \varphi)$

Sinusoidal quantities

Quantity	Symbol	Meaning / relation	Unit
Maximum value	V_m	Peak magnitude	V
Instantaneous value	v(t)	Value at a chosen time	V
Time period	T	Time for one cycle	s
Frequency	f	Cycles per second; f = 1/T	Hz
Angular velocity	ω	ω = 2πf	rad/s
RMS value	V_{rms}	$V_m/\sqrt{2}$ for sine wave	V

Worked example — 50 Hz supply

For f = 50 Hz: T = 1/50 = 0.02 s = 20 ms; ω = 2π×50 ≈ 314 rad/s. For V_{rms} = 230 V, $V_m = \sqrt{2} \times 230 \approx 325$ V.

Ohm’s Law calculator

Voltage V

24

Resistance Ω

12

Enter values and calculate.

Two-resistor parallel calculator

R_1 Ω

6

R_2 Ω

3

Enter values and calculate.

Module I summary: Current is charge flow; voltage is energy per charge; resistance controls current. Networks obey series/parallel rules. Changing magnetic flux induces emf. A sinusoidal AC waveform is described by peak, RMS, frequency, period and phase.

MODULE II

Domestic Electrical Supply System and Energy Utilization

Understand the route from service connection to final circuits, along with protection, earthing, wiring, shock prevention and bill calculation.

15 hours | CO2 | Remember · Understand · Apply

Official Module II subtopic allocation		
Subtopic	**Hours**	**Taxonomy**
Basic terms in electrical circuits	2	Understand
Domestic supply structure	2	Understand
Protection devices	3	Remember
Earthing	2	Understand
Electrical safety and shock	1	Understand
Residential wiring	2	Remember
Electrical energy consumption	3	Apply
Total	**15**	**CO2**

2.1 Basic circuit terms and single-/three-phase supply−

Phase / line

Live conductor carrying alternating potential relative to neutral and earth.

Neutral

Current-return conductor connected to the source star point and earthed at the supply system.

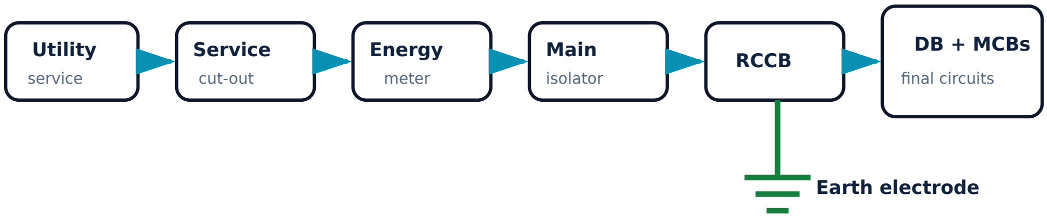
Protective earth

Safety conductor bonding exposed conductive parts to earth; it is not a normal load-current path.

A **single-phase** supply is commonly used for ordinary household loads. A **three-phase** supply has three phase conductors displaced by 120 electrical degrees and is suited to larger loads and motors.

> **Malayalam support:** Neutral-□□□ Earth-□□□ □□□ purpose □□□□. Neutral normal current return path □□□. Earth fault □□□□□□□ safety path □□□ □□□□□□□□□□□□□□□□□□. □□□□ load end-□ □□□□□□□□□□□□□ □□□□□□□□□□□.

2.2 Structure of a domestic electrical supply system−



Typical functional sequence. Exact utility and statutory arrangements must follow local regulations and competent-person practice.

The service connection brings supply to the premises. The energy meter records consumption. The main switch/isolator disconnects the installation. The distribution board divides supply into protected final circuits.

2.3 Protection devices — fuse, MCB, ELCB/RCCB and neutral link−

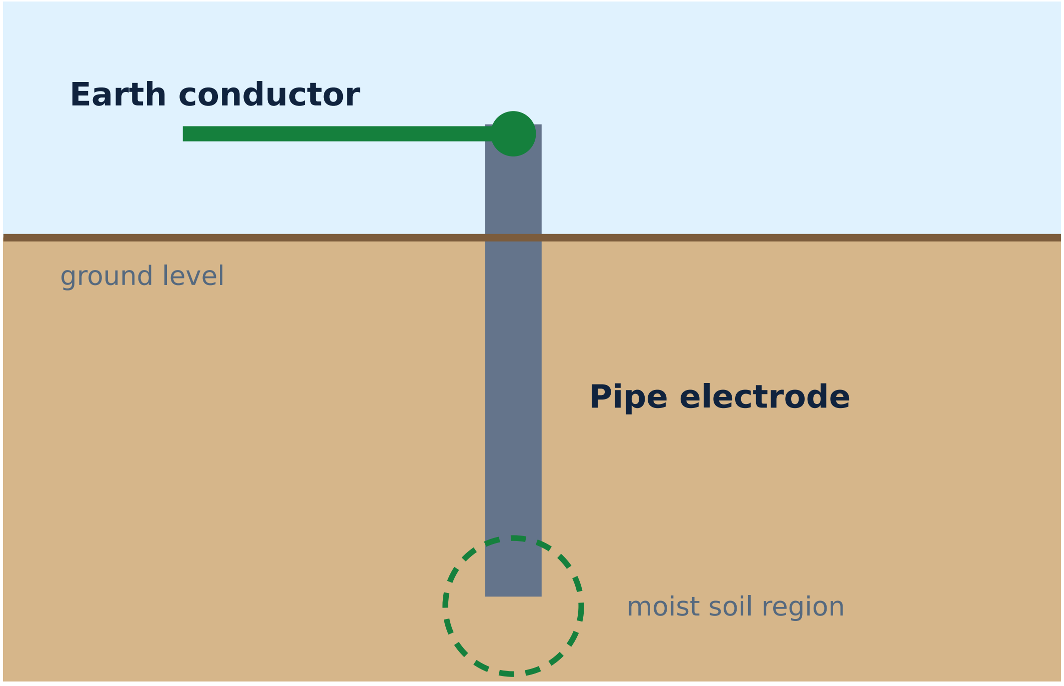
Protection devices and functions

Device	Purpose	Important point
Rewirable fuse	Melting fuse element opens on excessive current.	Correct fuse wire rating is essential; never bypass.
Cartridge fuse	Enclosed calibrated fuse element.	Better containment and predictable operation.
MCB	Trips for overload and short circuit.	Resettable after fault is cleared.
ELCB / RCCB	Detects earth leakage/residual current.	It does not replace overcurrent protection unless combined as RCBO.
Neutral link	Common termination/distribution of neutrals.	Must be correctly segregated, especially across RCCB circuits.

Safety: Protective devices are not permission to work live. Isolation, lockout, test-before-touch and competent supervision remain necessary.

2.4 Earthing — need, plate and pipe methods−

Earthing limits touch voltage and provides a low-impedance fault path so protective devices operate quickly. Main materials include the earth continuity conductor, earthing lead and earth electrode.



Conceptual pipe earthing arrangement; dimensions and construction must follow approved standards and local practice.

Plate vs pipe earthing

- **Plate:** metal plate electrode buried in ground.
- **Pipe:** perforated GI pipe used as electrode; widely used where suitable.

Soil condition, corrosion, moisture and electrode geometry influence earth resistance.

> **Malayalam support:** Earthing-□□□□ □□□□□□□ fault current safely □□□□□□□□□□□ □□□□□□□□□□ MCB/RCCB □□□□□□□□ operate □□□□□□□□□□□□ exposed metal □□□□□□□□□□ touch voltage □□□□□□□□□□□□□□□□□□□.

2.5 Electrical safety and electrical shock−

Shock severity depends on current path, magnitude, duration, frequency, skin condition and environment. Prevention includes insulation, enclosures, earthing, residual-current protection, correct overcurrent protection, dry work conditions, safe isolation and competent maintenance.

> **Emergency principle:** Do not touch a person who is still in contact with live electricity. Isolate supply first using a safe means, call emergency services and

provide first aid/CPR only when trained and the scene is safe.

Safe work sequence	
Step	**Action**
1	Identify all energy sources and obtain authorization.
2	Switch off and isolate.
3	Lock/tag where applicable.
4	Prove tester, test for dead, re-prove tester.
5	Earth/discharge stored energy where required.
6	Use correct PPE, tools and barriers.

2.6 Residential wiring systems and materials−

Wiring systems in the syllabus		
System	**Construction**	**Typical feature**
Cleat wiring	Insulated conductors supported on cleats.	Temporary and inexpensive; low mechanical protection.
Batten wiring	Sheathed cable fixed on wooden batten with clips.	Simple surface wiring.
Casing and capping	Conductors enclosed in casing with removable capping.	Neat surface route; material/environment limitations.
Open conduit	Conduit mounted on surface.	Good protection and accessible modification.
Concealed conduit	Conduit embedded in wall/ceiling.	Neat appearance; requires planning and workmanship.

Materials include conductors/cables, switches, sockets, ceiling roses/connectors, junction boxes, conduits, bends, saddles, distribution boards, protective devices and earthing accessories.

Design note: Cable size and protective-device rating must be selected from load current, installation method, voltage drop, short-circuit withstand, ambient conditions and applicable standards. Do not size a circuit from power alone.

2.7 Electrical energy consumption and monthly bill estimation−

Energy (kWh) = Power (kW) × Time (h)

The basic SI unit is joule. The commercial unit is kilowatt-hour: 1 kWh = 3.6×10^6 J.

Worked example — household energy

Four 10 W lamps operate 5 h/day, one 80 W fan operates 8 h/day, and a 1.5 kW heater operates 0.5 h/day for 30 days.

Lamps: 4×0.01×5×30 = 6 kWh. Fan: 0.08×8×30 = 19.2 kWh. Heater: 1.5×0.5×30 = 22.5 kWh.

Total = 47.7 kWh. At an assumed energy charge of ₹6 per kWh, energy charge only = ₹286.20.

Actual bills may include slabs, fixed charges, duty, subsidy, taxes and other items. Use the current tariff for a real bill.

Malayalam support: Appliance rating W-□ □□□□□□□□□ □□□□□ 1000 □□□□□□□ divide □□□□□□□ kW □□□□□□□. □□□□□□□ daily hours × days □□□□□□□ multiply □□□□□□□□. Bill calculation-□ current tariff slabs □□□□.

Energy-use estimator

Power (W)

1000

Hours/day

2

Days

30

Assumed rate (₹/kWh)

6

Energy and simple charge estimate will appear here.

Module II summary: A domestic installation needs correct isolation, metering, distribution, overcurrent protection, residual-current protection, earthing and sound wiring. Safety depends on design, maintenance and disciplined work practices—not on a single device.

MODULE III

Passive Electronic Components

Evolution of electronics, passive versus active components, resistor and capacitor coding, inductors and transformers.

15 hours | CO3 | Understand · Apply

Official Module III subtopic allocation

Subtopic	Hours	Taxonomy
Evolution of electronics	1	Understand
Resistor	2	Understand
Resistor colour coding	2	Understand
Capacitors	2	Understand
Capacitor number coding	2	Apply
Inductor	3	Understand
Transformer	3	Understand
Total	**15**	**CO3**

3.1 Evolution of electronics and passive/active classification−

Electronics moved from bulky vacuum tubes to semiconductor diodes and transistors, then integrated circuits (ICs), large-scale integration and VLSI. Integration reduced size and cost while increasing speed, reliability and functional density.

Passive and active components

Class	Meaning	Examples
Passive	Cannot provide power gain; stores, dissipates or transfers energy.	Resistor, capacitor, inductor, transformer.
Active	Can control current/voltage and may provide gain when powered.	Transistor, IC; semiconductor diodes perform active circuit functions though classification conventions vary.

Malayalam support: Passive component energy generate □□□□□□□□□□□; dissipate/store/transfer □□□□□□□□. Active component external supply □□□□□□□□□□□ signal control □□□□□□□□□□□ amplification □□□□□□□□.

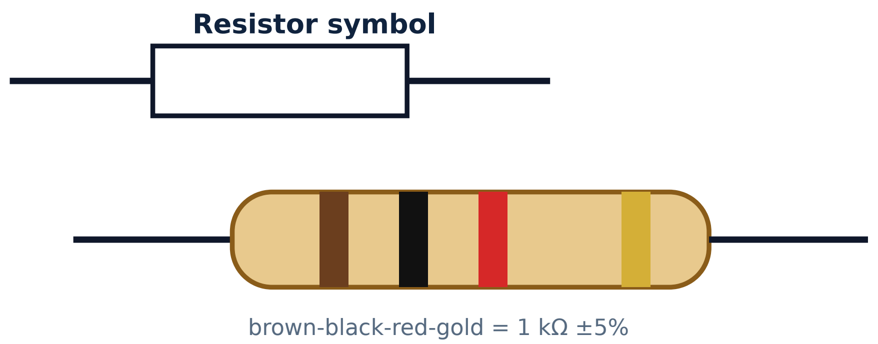
3.2 Resistor — symbol, function, specifications and types−

Main specifications

- Nominal resistance value (Ω)
- Tolerance (%)
- Power rating (W)
- Temperature coefficient where relevant

Types

- Fixed: carbon film, metal film, wire-wound
- Variable: potentiometer, rheostat, preset
- Special purpose: LDR, thermistor, varistor



A resistor must be selected for value, tolerance and power—not value alone.

3.3 Resistor colour code−

Standard digit and multiplier colours			
Colour	**Digit**	**Multiplier**	**Common tolerance**
Black	0	10^0	—
Brown	1	10^1	±1%
Red	2	10^2	±2%
Orange	3	10^3	—
Yellow	4	10^4	—
Green	5	10^5	±0.5%
Blue	6	10^6	±0.25%
Violet	7	10^7	±0.1%
Grey	8	10^8	±0.05%
White	9	10^9	—
Gold	—	10^{-1}	±5%
Silver	—	10^{-2}	±10%

Example

Brown–Black–Red–Gold: first digits 1 and 0, multiplier 10^2, tolerance ±5%.

$$10 \times 10^2 = 1000\ \Omega = 1\ \mathrm{k}\Omega \pm 5\%$$

3.4 Capacitor — capacitance, symbol, function, specifications and types−

A capacitor stores electric charge and energy in an electric field.

$$C = Q/V \qquad E = \tfrac{1}{2}CV^2$$

Capacitance is measured in farad (F), commonly μF, nF and pF. Common specifications include nominal capacitance, tolerance, rated voltage, dielectric type, temperature range and equivalent series resistance. The PDF also explicitly lists voltage, current and power ratings; their practical relevance depends on capacitor type and application.

Fixed

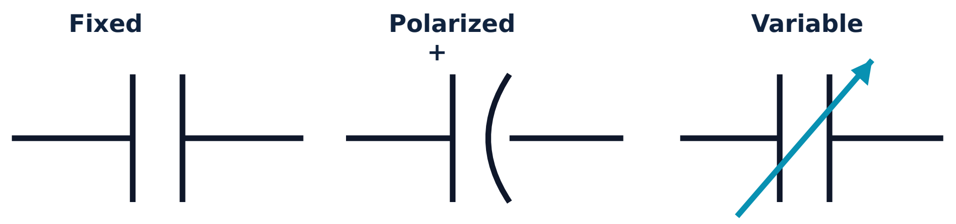
Ceramic, film, electrolytic, mica.

Variable

Air-variable and trimmer capacitors.

Special-purpose

Motor-run/start, pulse, safety-rated and supercapacitors.



Capacitor symbols: parallel plates for fixed, curved/marked polarity for polarized, and an arrow for variable capacitance.

Polarity: Electrolytic capacitors are usually polarized. Reversing polarity or exceeding rated voltage can cause failure. Capacitors can retain charge after power is removed.

Malayalam support: Capacitor charge store □□□□□□□. Electrolytic capacitor-□□□□ polarity □□□□□□□□□□□ damage □□□□□□□□□□ □□□□□ □□□□□□□□. Power off □□□□□ □□□□□□ charge □□□□□□□□.

3.5 Capacitor number coding−

For a common three-digit code, first two digits are significant figures and the third is the number of zeros in picofarads.

Examples

- **104:** 10×10^4 pF = 100,000 pF = 100 nF = 0.1 µF.
- **472:** 47×10^2 pF = 4700 pF = 4.7 nF.
- **221:** 22×10^1 pF = 220 pF.

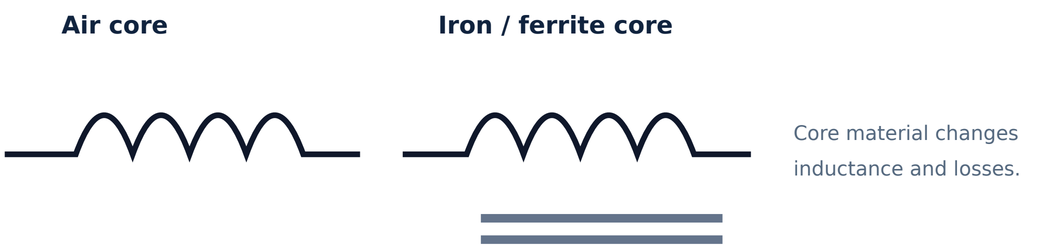
Some capacitors use direct values, letter tolerances or manufacturer-specific markings. Always inspect the full marking and datasheet.

3.6 Inductor — inductance, function, specifications and types−

An inductor stores energy in a magnetic field and opposes change in current.

$v = L\,di/dt \quad E = \frac{1}{2}LI^2$

Inductance is measured in henry (H). Types in the syllabus: air-core, iron-core and ferrite-core. Important practical parameters include inductance, tolerance, rated current, saturation current, winding resistance and core losses. The official syllabus also names nominal value, voltage rating, current rating and power rating.



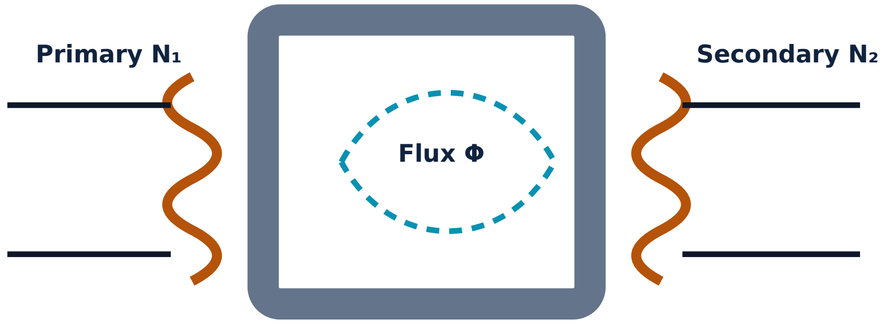
An air-core inductor has no magnetic core lines; iron/ferrite core is shown by parallel lines near the coil.

Malayalam support: Inductor current □□□□□□□□□□ change □□□□□□□□□□□ oppose □□□□□□□□. Capacitor voltage change oppose □□□□□□□□□□□□□□□□□□□ □□□□□□□□□□□□□□□; □□□□□□□ energy store □□□□□□□□□□□□ passive components □□□.

3.7 Transformer — self-induction, mutual induction and operation−

Self-induction is emf induced in the same coil by change in its current. **Mutual induction** is emf induced in one coil by changing current in another magnetically coupled coil.

$V_1/V_2 = N_1/N_2 = I_2/I_1$ (ideal transformer)



An AC primary current creates alternating core flux, inducing secondary voltage by mutual induction.

Step-up

$N_2 > N_1$, so $V_2 > V_1$ ideally.

Step-down

$N_2 < N_1$, so $V_2 < V_1$ ideally.

Worked example

A 230 V primary has 1000 turns; secondary has 100 turns.

$V_2 = V_1(N_2/N_1) = 230\times100/1000 = 23$ V.

Ideal secondary voltage = 23 V; it is a step-down transformer.

Resistor colour decoder (4-band)

Band 1

Brown 1

Band 2

Black 0

Multiplier

Red 2

Tolerance

Gold ±5%

Select bands.

Capacitor 3-digit decoder

Code

104

Enter a three-digit code.

Module III summary: Resistors dissipate energy, capacitors store electric-field energy, inductors store magnetic-field energy and transformers transfer AC energy through mutual induction. Correct component selection requires ratings as well as nominal value.

MODULE IV

Semiconductor Electronics and Digital Electronics Fundamentals

PN junctions, rectification, BJT action, number-system conversion, logic gates and recent electronics trends.

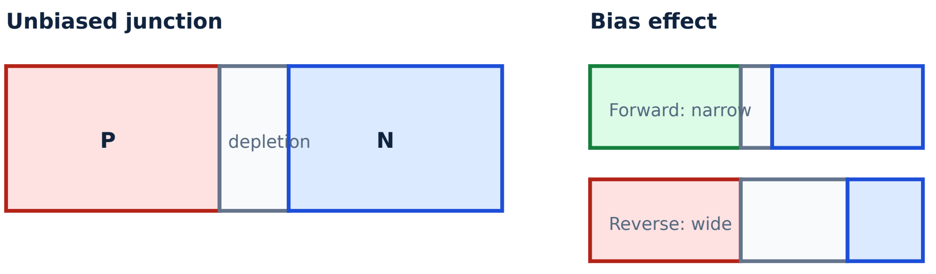
15 hours | CO4 (official mapping) | Remember · Understand · Apply

Official Module IV subtopic allocation		
Subtopic	**Hours**	**Taxonomy**
PN junction diode	2	Understand
Rectifier circuits	3	Understand
Bipolar Junction Transistor	2	Understand
Introduction to digital electronics	1	Understand
Number systems	4	Apply
Logic gates	2	Remember
Recent trends in electronics	1	Understand
Total	**15**	**CO4**

4.1 PN junction diode — structure, symbol, biasing and types−

Joining P-type and N-type semiconductor forms a depletion region and barrier potential. A diode conducts strongly in forward bias after sufficient junction voltage and blocks most current in reverse bias until breakdown.

Forward bias reduces the barrier; reverse bias increases it. Always observe polarity and ratings.



Forward bias narrows the depletion region; reverse bias widens it. Diagram is conceptual and not to scale.

Types include rectifier diode, signal diode, Zener diode, LED, photodiode, Schottky diode and varactor diode.

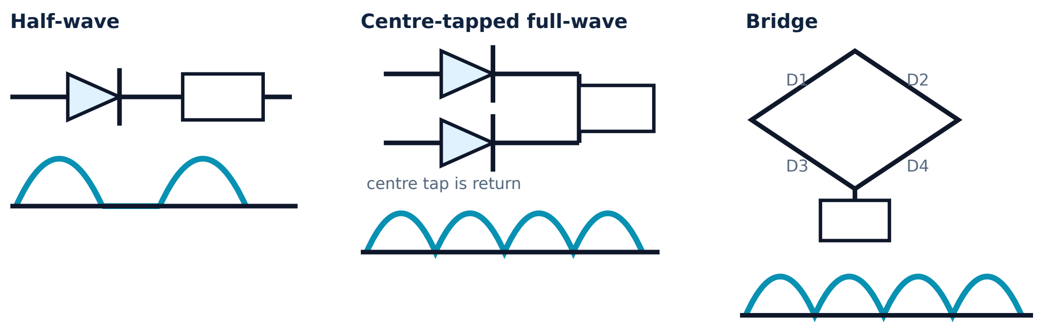
> **Malayalam support:** Forward bias-□ P side positive-□□□ N side negative-□□□ □□□□□□□□□□□□□□□□□□□□. Reverse bias-□ connection □□□□□□□□□□. LED-□□□□□ polarity □□□□□.

4.2 Rectifier circuits — half-wave, centre-tapped full-wave and bridge−

A rectifier converts AC into unidirectional pulsating DC. A filter can smooth the pulsation, but filter design is outside the listed detailed topics.

Rectifier comparison

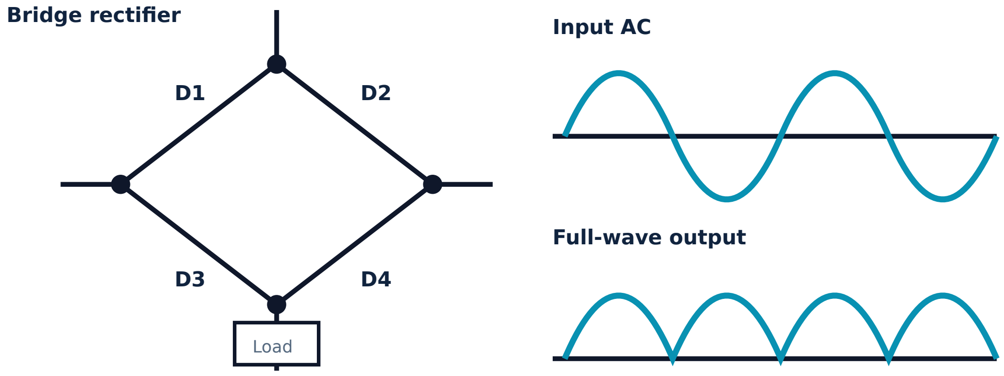
Type	Diodes	Conduction	Ripple frequency	Main feature
Half-wave	1	One half-cycle	f	Simple, poor utilization
Full-wave centre-tapped	2	Alternate diode each half-cycle	2f	Requires centre-tapped transformer
Bridge	4	Two diodes conduct each half-cycle	2f	No centre tap; common arrangement



Half-wave uses one half-cycle; full-wave circuits produce two pulses per input cycle.

Circuit symbols are simplified for learning. Draw complete source and diode polarity in examination answers.

Comparison of the three official rectifier circuits and output-waveform shapes.



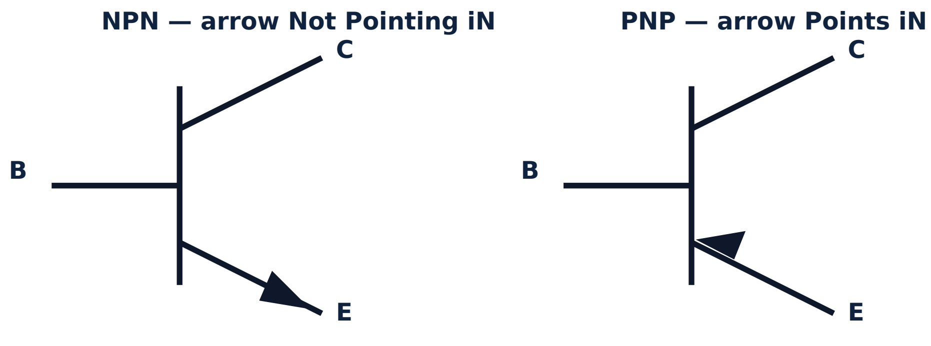
Bridge action makes load current flow in the same direction for both AC half-cycles.

Common mistake: In a bridge, two diode drops are in the conducting path. Current direction through the load remains unchanged although the active diode pair changes.

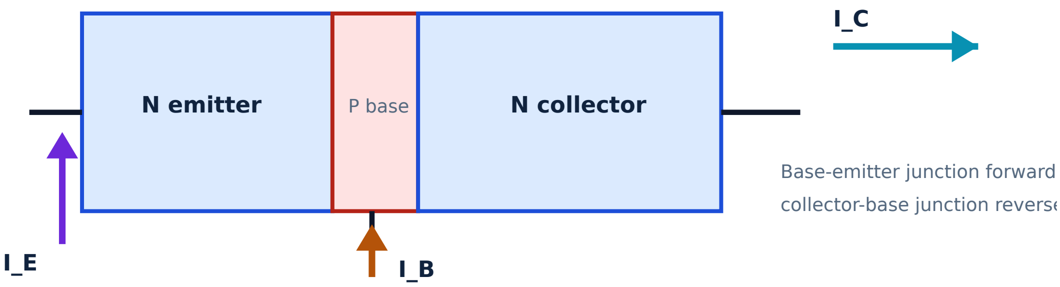
4.3 Bipolar Junction Transistor (BJT) — NPN, PNP and applications−

A BJT has emitter, base and collector regions. A small base current controls a larger collector current in common-emitter operation.

$I_E = I_C + I_B \quad \beta = I_C/I_B$



Arrow is on the emitter. NPN arrow points out; PNP arrow points in.



A thin base allows a small base current to control a larger collector current. I_E = I_C + I_B.

Applications include switching, amplification, oscillation, signal conditioning, relay/LED driving and power-control stages.

4.4 Introduction to digital electronics−

Digital systems represent information using discrete logic levels, commonly binary 0 and 1. Advantages include noise tolerance, repeatability, storage, programmability and easy integration. Applications include computers, controllers, communication, measurement, automation and consumer products.

> **Malayalam support:** Analog signal continuous value □□□□□□□□□. Digital system □□□□□□□□□□□ discrete 0/1 logic levels □□□□□□□□□□□□□□□. Real voltage ranges □□□ 0/1 represent □□□□□□□□□□□□; exact zero volt □□□□□□□ □□□□□□□□.

4.5 Number systems and conversions−

Common number systems

System	Base	Digits	Example
Decimal	10	0–9	245_{10}
Binary	2	0,1	101101_{2}
Octal	8	0–7	345_{8}
Hexadecimal	16	0–9, A–F	$2AF_{16}$

Binary to decimal

$101101_2 = 1\times2^5 + 0\times2^4 + 1\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0 = 32+8+4+1 = 45_{10}$.

Decimal to binary

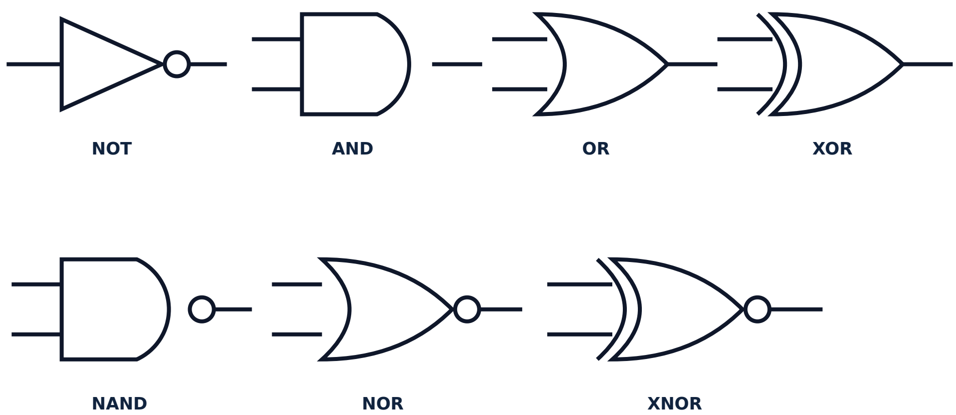
Repeatedly divide 45 by 2 and read remainders upward: $45_{10} = 101101_2$.

Hex and binary grouping

$2F_{16} \rightarrow 0010\ 1111_2$. Conversely $11010110_2 \rightarrow 1101\ 0110 \rightarrow D6_{16}$.

> Octal digits group binary in 3s; hexadecimal digits group binary in 4s. Pad zeros only on the left of an integer binary group.

4.6 Logic gates — symbols, equations, truth tables and uses−



A small output circle means inversion. XOR/XNOR have an extra curved input-side line.

Truth tables for two-input gates

A	B	AND A·B	OR A+B	XOR A⊕B	NAND $\overline{(A\cdot B)}$	NOR $\overline{(A+B)}$	XNOR $\overline{(A\oplus B)}$
0	0	0	0	0	1	1	1
0	1	0	1	1	1	0	0
1	0	0	1	1	1	0	0
1	1	1	1	0	0	0	1

NOT gate	
A	**Y = Ā**
0	1
1	0

AND

Output 1 only when all required conditions are 1. Application: interlocking/permissive logic.

OR

Output 1 when any input is 1. Application: multiple alarm sources.

XOR

Output 1 when inputs differ. Application: parity and half-adder sum.

NAND / NOR

Universal gates; any Boolean function can be built using only NANDs or only NORs.

XNOR

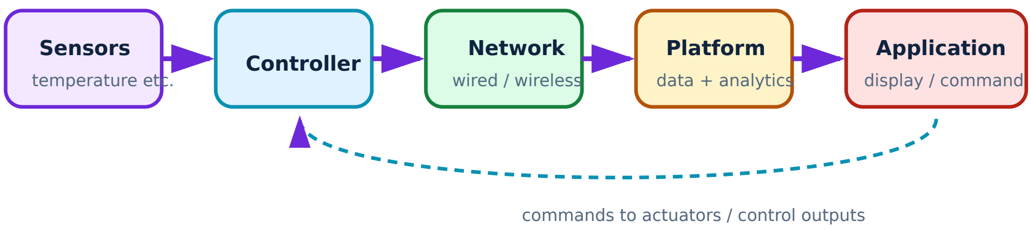
Output 1 when inputs are equal. Application: equality comparison.

NOT

Inverts one input. Application: active-low signal conversion.

4.7 Recent trends — Internet of Things (IoT) and Artificial Intelligence−

IoT connects sensors, controllers, networks and applications so physical systems can be monitored and controlled. **AI** enables systems to perform tasks such as pattern recognition, prediction, classification and decision support using data and models.



IoT is a complete sensing–communication–application chain; AI may be used in the analytics stage.

Cyber and ethical note: Connected devices require authentication, updates, secure communication, data minimization and safe fail states. AI output must be validated before safety-critical use.

Number-system converter

Input value

45

Input base

Decimal

Conversions will appear here.

Module IV summary: A diode controls direction of current; rectifiers convert AC to pulsating DC; a BJT can switch or amplify. Digital systems use binary logic, while number conversion and truth tables are core analytical tools. IoT and AI extend electronics into connected, data-driven systems.

QUICK REFERENCE

Formula, Rule and Procedure Bank

Charge and current

$I = Q/t$

Voltage

$V = W/Q$

Ohm's Law

$V = IR$

Power

$P = VI = I^2R = V^2/R$

Resistance law

$R = \rho L/A$

Series resistance

$R_T = \Sigma R$

Parallel resistance

$1/R_T = \Sigma(1/R)$

Faraday law

$e = -N\, d\Phi/dt$

Sinusoid

$v = V_m \sin(\omega t+\varphi)$

Frequency

$f = 1/T;\ \omega = 2\pi f$

RMS sine value

$V_{rms} = V_m/\sqrt{2}$

Energy

$E(kWh)=P(kW)t(h)$

Capacitor

$C=Q/V;\ E=\frac{1}{2}CV^2$

Inductor

$v=L\,di/dt;\ E=\frac{1}{2}LI^2$

Transformer

$V_1/V_2=N_1/N_2$

BJT

$I_E = I_C + I_B$; $\beta = I_C / I_B$

Generator rule

Fleming's Right-Hand Rule: first finger → magnetic field, thumb → motion, second finger → induced current.

Motor rule

Fleming's Left-Hand Rule: first finger → field, second finger → current, thumb → force/motion.

VISUAL LIBRARY

Diagram Library for Rapid Revision

Electrical

AC/DC waveforms, resistor networks, generator and sinusoidal parameters.

Open diagrams

Domestic supply

Supply chain, earthing and wiring-system comparison.

Open diagrams

Electronics

Resistor bands, transformer, diode, bridge rectifier, BJT and IoT.

Open diagrams

SELF-LEARNING CENTRE

Official Suggested Student Activities

The official table assigns **6 hours to each activity** and totals **90 self-learning hours**. Activities below retain the official intent and add a safe submission format.

1 · Extension board

Sketch a home extension board and explain why connected devices receive the same voltage: sockets are wired in parallel.

Submit: labelled sketch, parallel-path explanation, safety observations.

2 · Series divider

Draw three series resistors with assumed values and calculate each voltage using voltage division.

Check: sum of drops equals source voltage.

3 · AC/DC appliance survey

List household appliances, AC/DC operation and power rating.

Safety: read nameplates only; do not open energized equipment.

4 · Fleming's Left-Hand Rule

Prepare a short note on finger meanings and applications. Officially listed under Module I.

Clarify: Left-Hand Rule is for motor force; Right-Hand Rule is for generator induction.

5 · Repeated Left-Hand Rule activity

The same official task appears again under Module II. Use this submission to compare Left- and Right-Hand Rules in a table.

6 · Home electrical safety

List safety measures: MCB/RCCB, earthing, enclosure, safe sockets, dry conditions, no overloading and periodic inspection.

7 · Monthly bill estimate

List major equipment, power ratings and usage. Compute kWh and estimate the bill using the current tariff.

8 · Five resistor codes

Identify nominal value and tolerance of five standard resistors.

Record: band sequence, calculation, measured value if supervised.

9 · Three capacitor codes

Decode three standard capacitor number markings.

10 · Ceiling-fan capacitor

Identify the capacitor type and explain its role in creating phase shift for starting/running.

Safety: use nameplate/photo/reference; do not open or touch a live fan.

11 · Induction cooker

Explain how high-frequency magnetic field induces currents in compatible cookware and produces heat.

12 · Diode types

Create a feature-and-use table for rectifier, signal, Zener, LED, photodiode, Schottky and varactor diodes.

13 · Component inventory

Official task says open faulty equipment. Perform only under instructor supervision after safe isolation and capacitor discharge.

14 · Three-digit conversion

Choose a three-digit decimal number and convert it to binary, octal and hexadecimal. Verify by reconversion.

15 · Recent product review

Choose a newly released electronic product and describe features and technology. Include source date and distinguish marketing claims from verified specifications.

OFFICIAL ASSESSMENT

Assessment Methodology and Examination Pattern

CIE and ESE structure			
Component	**Official detail**	**Converted / final marks**	**Tentative schedule**
Attendance	End-of-semester attendance component	5	End of semester
CA1–CA4 self-learning	Module 1, 2, 3 and 4 activities; each row shows exam mark 15	Combined self-learning contribution 15	By 4th, 7th, 11th and 14th weeks
CA5 and CA6 series exams	Each 2 h, 50 marks, two modules	Combined series contribution 20	8th and 15th weeks
ESE	Written theory, 2.5 h, 60 marks	60	End of semester
Total		**100**	**—**

Series exam pattern — 2 hours, 50 marks

Any two modules. For each module: Part A has 2 one-mark questions, Part B has 3 three-mark questions, and Part C has 2 seven-mark questions with choice, giving 25 marks per module. Overall question counts shown: Part A 4, Part B 6, Part C 4.

ESE pattern — 2.5 hours, 60 marks

Each module contributes 15 marks: Part A 2×1 = 2; Part B answer 2 of 3, 2×3 = 6; Part C one set with choice = 7. Across four modules: 8 + 24 + 28 = 60.

EXPECTED / PRACTICE QUESTIONS

Module-wise Questions with Answers

These are practice questions aligned to the official syllabus. They are not claimed as official SITTTR questions.

Module I

Define current, voltage and resistance with SI units.−

Current is charge flow rate $I=Q/t$, unit ampere. Voltage is energy/work per charge $V=W/Q$, unit volt. Resistance is opposition to current $R=V/I$, unit ohm.

A 20 Ω resistor is connected to 100 V. Find current and power.−

$I=V/R=100/20=5$ A. $P=VI=100\times5=500$ W.

Three resistors 2 Ω, 3 Ω and 6 Ω are in parallel. Find equivalent resistance.−

$1/R=1/2+1/3+1/6=1$, so $R=1\ \Omega$.

State Faraday's laws and explain the negative sign in $e=-N\, d\Phi/dt$.−

First law: changing linked magnetic flux induces emf. Second law: induced emf magnitude is proportional to rate of change of flux linkage. Negative sign expresses Lenz's law—induced action opposes the change that produces it.

For a 50 Hz sine wave, find period and angular velocity.−

$T=1/f=0.02$ s$=20$ ms. $\omega=2\pi f\approx314$ rad/s.

Module II

Differentiate phase, neutral and earth.−

Phase is the live supply conductor. Neutral is the normal return conductor connected to the source reference/star point. Protective earth bonds exposed metal to earth for fault protection and is not a normal load-current conductor.

Compare MCB and RCCB.−

MCB protects against overload and short circuit. RCCB detects imbalance/residual current indicating earth leakage. An RCCB alone does not provide conventional overcurrent protection unless combined in an RCBO.

Why is earthing necessary?−

It limits touch voltage and provides a fault-current path so protective devices operate quickly, reducing shock and fire risk.

A 2 kW heater runs 3 h/day for 30 days. Find energy.−

$E=2\times3\times30=180$ kWh.

List residential wiring systems in the syllabus.−

Cleat, batten, casing-and-capping, open conduit and concealed conduit wiring.

Module III

Distinguish passive and active components.−

Passive components do not provide power gain and mainly dissipate, store or transfer energy. Active components can control signals/current and may provide gain when powered.

Decode brown-black-red-gold.−

Digits 1,0; multiplier 10^2; tolerance ±5%. Value=$10\times100=1000\ \Omega=1\ k\Omega$ ±5%.

Decode capacitor code 472.−

47×10^2 pF=4700 pF=4.7 nF.

Differentiate self-induction and mutual induction.−

Self-induction is emf induced in the same coil due to change of its own current. Mutual induction is emf induced in a second coil due to changing current/flux from the first.

A transformer has N_1=500, N_2=100 and V_1=230 V. Find V_2.−

$V_2=V_1N_2/N_1=230\times100/500=46$ V. It is step-down.

Module IV

Explain forward and reverse bias of a PN diode.−

Forward bias connects P to positive and N to negative, reducing junction barrier and allowing current. Reverse bias connects P to negative and N to positive, increasing barrier and allowing only small leakage until breakdown.

Why does a full-wave rectifier have output ripple frequency 2f?−

Both positive and negative AC half-cycles are converted into positive load pulses, producing two output pulses per input cycle.

State the BJT current relation and one application.−

$I_E=I_C+I_B$. Applications include switching, amplification and relay driving.

Convert 101101_2 to decimal and hexadecimal.−

Decimal=45. Group as 0010 1101, so hexadecimal=$2D_{16}$.

Give the XOR and XNOR condition.−

XOR output is 1 when inputs differ. XNOR output is 1 when inputs are equal.

EXAMINATION PRACTICE

Practice Model Paper — based on the official pattern, not an official SITTTR model question paper

Duration: 2.5 hours **Maximum:** 60 marks

Part A — 8 × 1 = 8

1. State the SI unit of resistivity.

2. Define RMS value.
3. Name one residual-current protection device.
4. State the commercial unit of electrical energy.
5. What does the third digit in capacitor code 104 represent?
6. Define mutual induction.
7. Write the equation of a NOT gate.
8. What is the base of hexadecimal?

Part B — answer 2 of 3 from each module; 12 × 3 = 36 offered, 24 marks answered

Module I: Explain AC vs DC; solve a voltage divider; state Faraday’s laws.

Module II: Compare fuse and MCB; explain pipe earthing; calculate a simple monthly energy value.

Module III: Explain resistor specifications; decode capacitor codes; differentiate step-up and step-down transformers.

Module IV: Explain diode biasing; convert decimal to binary/hex; draw AND, OR and XOR truth tables.

Part C — one 7-mark set with choice from each module; 4 × 7 = 28

1. **Module I:** Solve a series–parallel circuit with voltage/current division *or* explain AC generation and sinusoidal parameters with diagram.
2. **Module II:** Draw and explain domestic supply with protection and earthing *or* compare residential wiring systems and safety practices.
3. **Module III:** Explain R, C and L with types/specifications *or* explain transformer construction, principle and numerical ratio problem.
4. **Module IV:** Explain three rectifiers with waveforms *or* explain BJT and digital logic gates with truth tables.

ONE-LOOK REVISION

Quick Revision, Common Mistakes and Exam Checklist

Module I

$V=IR$; series adds; parallel reciprocal; $e=-N\, d\Phi/dt$; $f=1/T$; $RMS=peak/\sqrt{2}$.

Module II

Supply → meter → isolator → RCCB/DB; MCB for overcurrent; earthing for fault safety; kWh=P×t.

Module III

R dissipates, C stores electric energy, L stores magnetic energy, transformer uses mutual induction.

Module IV

Diode direction, full-wave uses both halves, NPN arrow out, number bases 2/8/10/16, truth tables.

Common mistakes

- Confusing electron flow with conventional current.
- Using current division formula for a series circuit.
- Adding parallel resistances directly.
- Confusing neutral with protective earth.
- Ignoring component tolerance, polarity or power/voltage rating.
- Using Left-Hand Rule for generator direction.
- Grouping binary incorrectly for octal/hex conversion.

Exam checklist

- Write formula before substitution.
- Convert prefixes and units correctly.
- Draw clear labels and current/polarity arrows.
- Show truth-table headings and logic equation.
- State assumptions for ideal transformer or simple tariff.
- For safety answers, include isolation, protection and earthing.

TERMINOLOGY

Glossary

Important terms	
Term	**Meaning**
Charge	Electrical property measured in coulomb.
Potential difference	Energy transferred per unit charge.
Resistivity	Material property relating resistance to geometry.
RMS	Effective AC value producing the same heating as an equivalent DC value.
Electromagnetic induction	Production of emf by changing magnetic flux linkage.
Residual current	Difference between outgoing and returning current, indicating leakage.
Capacitance	Charge stored per volt.
Inductance	Property opposing change of current through induced emf.
Rectification	Conversion of AC to unidirectional pulsating current.
Logic level	Voltage range interpreted as digital 0 or 1.
IoT	Connected sensing, control and data-processing system.
AI	Data/model-based methods for tasks such as prediction or recognition.

LEARNING RESOURCES

Official References

Textbooks

1. B. L. Theraja and A. K. Theraja, *Textbook of Electrical Technology: Part 1 — Basic Electrical Engineering in S. I. Units*, S. Chand Publication.
2. J. B. Gupta, *Electronic Devices and Circuits*, Katson Books.

References

1. N. N. Bhargava, D. C. Kulshreshtha and S. C. Gupta, *Basic Electronics and Linear Circuits*, Tata McGraw Hill Education.
2. A. Anand Kumar, *Fundamentals of Digital Circuits*, Prentice Hall India Pvt. Ltd.

Officially listed web resources

- NPTEL — Basic Electrical Technology, Course 108108076 (Modules I & II).
- NPTEL — Fundamentals of Electrical Engineering, Course 108105112 (Modules I & II).
- NPTEL — Basic Electronic Course, Course 122106025 (Module III).
- NPTEL — Digital Electronic Circuits, Course 108105132 (Module IV).